Research article

Forecasting Nepal's Electricity Demand under Transport and Cooking Electrification Using Deep Learning

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Abstract

The goal of this research was to develop an analytical technique to assess the growing use of power in Nepal's transportation and cooking sectors. To achieve this, a two-step LSTM model was developed that captures both the economic factors affecting energy consumption in Nepal as well as the new demands on electricity created by increased use of electric vehicles (EVs) and electric cooking appliances. The first step used univariate LSTM models to produce forecasts of demographic and economic data, including gross domestic product (GDP), population, urban development, and access to electricity. Simultaneously, estimates of energy consumption by EVs and electric cookers were developed based on analyses of historical trends and planning documents under Nepal's Nationally Determined Contributions (NDC). In the second step, a multivariate LSTM model was created to establish a benchmark for electricity consumption by separating the effects of electrification from historical consumption data. Once the benchmark for electricity consumption was established, the projected EV and electric cooking energy use were added to the baseline to estimate total demand. The results indicate that electricity demand in Nepal is projected to grow from 16.42 TWh in 2026 to 28.77 TWh in 2030, with approximately 17-18% of total demand driven by electrification. The results demonstrate the need for coordinated electricity planning and development, including expanding infrastructure and implementing demand-side management strategies.

1. Introduction

The global energy sector is undergoing a fundamental transformation away from traditional fossil fuels to cleaner renewable sources of energy to mitigate the impacts of climate change and other environmental concerns [1], [2]. The United Nations has established Sustainable Development Goal Number Seven (SDG 7) as one of their 2030 Agenda goals for achieving universal access to affordable, reliable, sustainable and modern energy by 2030 [3]. Developing countries will experience this transition to modern energy on two levels: first, they will begin transitioning their transportation systems from dependence upon imported fuels to electric vehicles; and second, these countries will be transitioning away from using traditional biomass fuels for cooking in favor of new modern energy cooking options [4], [5]. The two sets of modern energy described above are essential in addressing the impacts of climate change as well as public health issues; however, integrating them into existing under-resourced and technically weak distribution networks may pose significant risks for grid reliability and stability [2], [4], [6].

Nepal's energy system stands apart from those of most other countries. It can generate up to 45.6 GW of clean hydropower, more than the technically feasible capacity needed to meet national demand. However, most of Nepal's constituents still use outdated methods to produce energy for their families [7]. Approximately 60% of the population uses solid biomass or other traditional energy for home consumption, and about 33% of households use imported LPG [7]. The Government of Nepal has set measures to promote energy independence and reduce capital outflows. By implementing aggressive programs to add an electric stove per household and establishing high targets by 2030, approximately 90% of private two-wheeled vehicles and 60% of public two-wheeled vehicles will be EVs [8], [9], [10]. A large number of electric stoves and EVs could result in significant load-shifting impacts across Nepal and substantial changes to Nepal's electrical system load profile [11]. Past research suggests that an uncoordinated EV charging will create substantial peak demand. Furthermore, widespread installation of electric cooking systems will cause distribution transformers to overload and as a result, create unacceptably low voltage levels at the end of distribution feeders [11], [12], [13].

Due to the stochastic nature, high-power capabilities and coincidental timing of EVs and electric cooking systems' peak loading and that of the existing peak electricity demand for lighting and entertainment and result in a new peak that may exceed the equipment capacity of distribution substations [8], [11], therefore brownout or blackouts of the electrical system are likely to occur unless aggregated EV charging impacts and electric cooking impacts to the grid are accurately forecasted and assessed by energy policymakers responsible for providing reliable electric infrastructure and minimizing mismatch between supply & demand [2], [14], [15].

Traditional forecasting methods generally fail to capture the complex, non-linear, and time-sensitive nature of these new loads [2], [16]. Therefore, modeling the estimated demand from aggregated electric vehicles and electric cooking systems requires new models that account for the range of consumer behaviors, the seasonality of energy use, and the uncertainty of “fuel stacking”

behaviors [4], [16], [17], [18]. Predictive analytics and machine learning are key tools for maintaining safe, reliable grid operation [2]. Among them, Long Short-Term Memory (LSTM) networks (a type of recurrent neural network (RNN) that is specifically designed to minimize its objective function and to accurately forecast energy demand) outperform other algorithms [2]. Some research demonstrates the performance of LSTM model with other models like ARIMA and SARIMA and the error in LSTM model was found extremely lower compared to them [19], [20]. Recent work has continued to extend these recurrent-network approaches for load forecasting. Comparative studies confirm that LSTM and bidirectional LSTM models outperform classical statistical methods such as ARIMA and SARIMA for nonlinear energy-consumption time series [19]. Beyond single-network designs, hybrid and multivariate architectures have become more common: hybrid convolutional–recurrent models (CNN-LSTM and CNN-GRU) that combine feature extraction with temporal modelling have been shown to improve robustness during peak-load intervals on multivariate national-grid datasets [21], and multivariable CNN-LSTM models that jointly ingest load and macroeconomic indicators have outperformed approaches trained on either signal alone [22]. Such evidence motivates the multivariate, macroeconomically driven LSTM design adopted here. Comparatively little of this work, however, addresses developing-country contexts, where short historical records and rapid structural change complicate forecasting; recent applications to settings such as Bangladesh and Iran demonstrate the feasibility of deep-learning load forecasting under these constraints [23], [24] and underscore the relevance of the present Nepal-focused study.

A multivariate LSTM model has been developed to predict electricity demand for the nation of Nepal that reflects the increasing penetration of EVs and electric cooking, by incorporating charging power loads from historical datasets, travel patterns, and household cooking diaries. The objective of this research is to produce prediction for long-term (i.e. yearly) changes in demand.

The outputs of this model will assist on data-driven roadmap for utility stakeholders, providing information necessary to make informed decisions on infrastructure upgrades, transformer resizing, and demand-side management strategies to facilitate a reliable grid during Nepal's transition to clean energy.

While previous studies have analyzed EV charging and electric cooking independently, very few studies integrate both within a unified forecasting framework, particularly in hydropower-dominated developing countries. Most existing studies also focus on annual energy demand, with limited attention to the temporal demand dynamics relevant for grid operation. This study addresses these gaps by combining electrification scenarios with deep learning-based demand forecasting.

2. Methodology

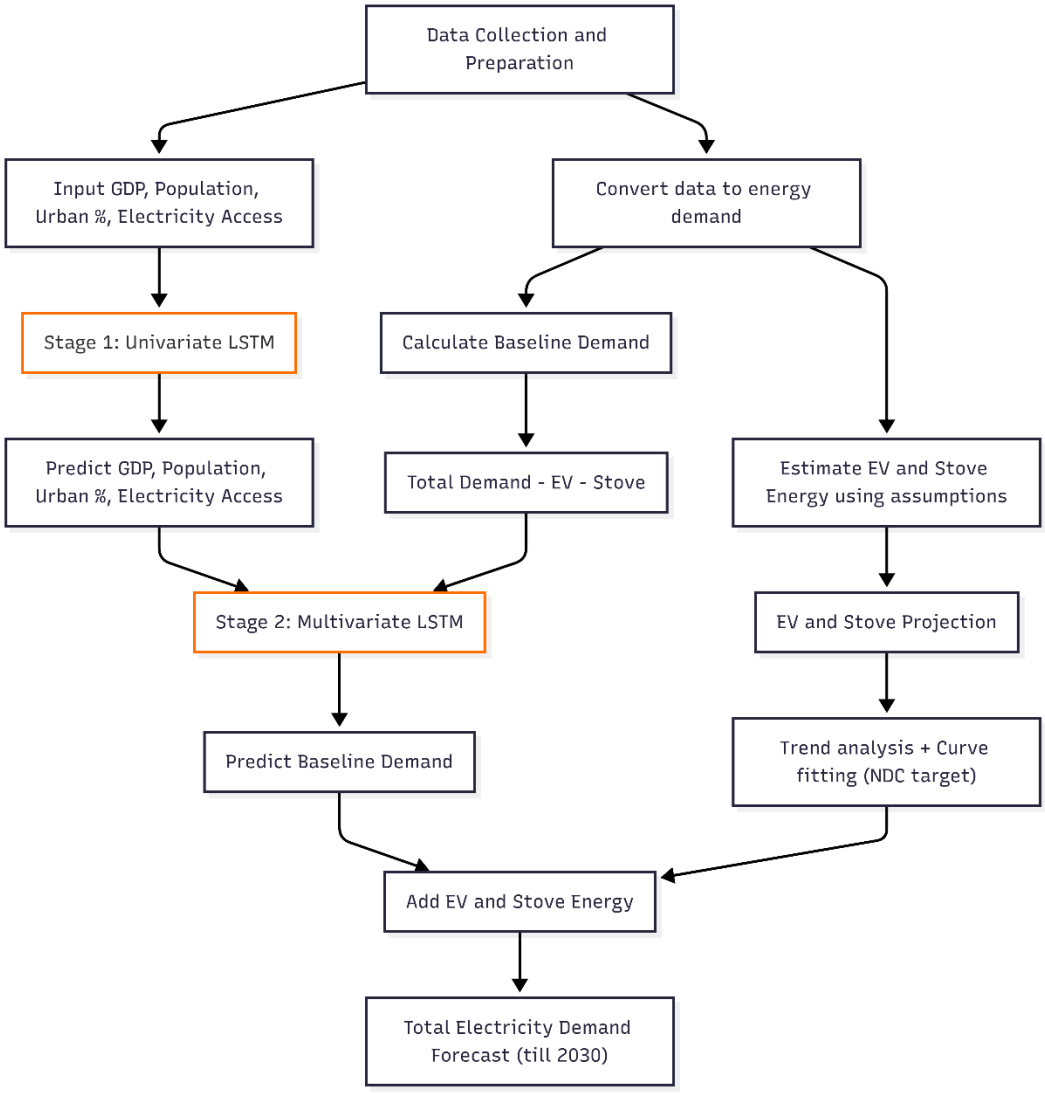


Figure 1: Methodology

The research begins with data collection and processing. GDP, population, urban population percentage, energy consumption per capita and electricity access were taken as input parameters. The first stage of forecasting was done with univariate LSTM for GDP, population, urban population percentage, electricity consumption per capita electricity consumption till 2030. Those data along with other parameters like baseline energy and electricity access is passed to multivariate LSTM to project future electricity demand. The baseline forecast was then added to projected energy demand for EVs and stoves. Figure 1 illustrates the overall methodological workflow, including data preprocessing, feature forecasting using univariate LSTM models, baseline demand estimation using multivariate LSTM, and integration of electrification scenarios. A feedback loop is incorporated to refine model performance.

2.1. Obtaining and Processing Data

The data used in this research was derived from secondary sources, i.e. from national energy reports from NEA and WECS, NDC targets and from other international databases and information sources [25], [26], [27], [28], [29], and organized into a time-series database of Nepal's demand for electricity, as well as on associated or complementary socio-economic measurements. Variables were included for total demand for electricity, Gross Domestic Product (GDP), total population, percentage of urban population, percentage of people with access to electricity, electric vehicles (EV) energy consumption and electric cooking technologies.

After cleaning the data, the data were organized chronologically. The final time-series was consolidated into a single variable that measured total annual cumulative electricity consumption in terawatt-hours (TWh). In order to improve the convergence of the models and the numerical stability of the models, all of the inputs to the models were normalized using the Min-Max scaling technique.

2.2. Converting Energy Demand and Platforms

In order to characterize the energy consumption of different categories of electric vehicles considered in this study, Table 1 presents the annual energy consumption, annual distance travelled, and the corresponding efficiencies expressed in km/kWh. For instance, two-wheel electric vehicles consume approximately 295.42 kWh/year and travel an average distance of 8,952 km [30], resulting in an efficiency of 3.3 km/kWh [31]. Similarly, three-wheel electric vehicles consume 1,098 kWh/year with an annual travel distance of 18,000 km [32], corresponding to an efficiency of 6.1 km/kWh [31]. Four-wheel electric vehicles consume approximately 1,514.13 kWh/year and travel 12,310 km annually [30], yielding an efficiency of 12.3 km/kWh [31]. Public transportation buses exhibit significantly higher energy usage, consuming 60,073.28 kWh/year while travelling 58,112 km [33], resulting in an efficiency of 103.38 km/kWh [33]. Lastly, commercial electric vehicles consume approximately 3,690 kWh/year and travel 30,000 km annually [30], corresponding to an efficiency of 12.3 km/kWh [31]. These values are used to estimate segment-wise electricity demand, allowing the contribution of electric mobility to be incorporated into the overall electricity demand forecasting framework.

Table 1: Annual Energy consumed by different vehicle categories

Vehicle Category	Annual Energy (kWh)	Annual Distance (km)	Efficiency (km/kWh)
Two-wheelers	295.42	8,952	3.3
Three-wheelers	1,098	18,000	6.1
Four-wheelers	1,514.13	12,310	12.3
Buses	60,073.28	58,112	103.38

Commercial	3,690	30,000	12.3

These values reflect refined vehicle-usage assumptions and efficiency estimates, giving a more realistic estimate of sectoral electricity demand.

The energy consumption values for electric vehicles are derived from representative operational statistics based on annual mileage and efficiency estimates. Similarly, the induction cooking energy requirement (approximately 2.36 kWh/day) is calculated using LPG-to-electric energy equivalence based on thermal efficiency differences between LPG and induction systems. The literature indicates that one 14.2 kg LPG cylinder is consumed in 41.9 days [34], with average thermal efficiencies of 48.27% for LPG cooking and 88.62% for induction cooking [35], the equivalent induction electricity consumption is estimated at 2.36 kWh/day.

While these assumptions provide a reasonable approximation, actual consumption may vary depending on user behavior and appliance efficiency. Therefore, the results should be interpreted as scenario-based estimates rather than exact predictions.

2.3. Two-Stage LSTM Modeling Framework

The forecast for electricity demand was done by placing forecasting into two different stages with a LSTM forecasting framework. These two stages are; forecasting macroeconomic drivers of electricity demand and estimating the baseline demand and incorporating electrification loads. This approach facilitates more interpretive modelling due to the separate modelling of future structural demand versus future electrification trends.

2.4. Feature Forecasting and Electrification Modeling

In the first stage, univariate LSTM models were created to formulate the four driver variables used in the modelling process: GDP, Total Population, Percentage of Urban Population, and Electricity Access Rate. Each of these variables is modelled utilizing a time-series data method, where the sliding-window approach is applied to the construction of each model. The selection of LSTM networks as a method of modelling is primarily due to their ability to capture and store non-linear temporal dependencies between observations and models through long-term memory.

This model predicts future values for individual macro-parameters based on a lookback window of 2 years.

- Input Features: 1 (GDP, Population, Per Capita Energy Consumption)
- Hidden Units: 32
- Architecture:
    - LSTM Layer: 1 layer with 32 hidden units

    - Fully Connected Layer 1: Linear (32 → 16)
    - Activation: ReLU
    - Fully Connected Layer 2: Linear (16 → 1)
- Optimizer: Adam (lr=0.005)
- Training Epochs: 250

In parallel to the first stage, energy demand from electric vehicles and electric cooking was forecasted using a hybrid approach. Historical demand trends were used to forecast future demand through the use of curve-fitting techniques, which produced estimates in alignment with the Nationally Determined Contributions of Nepal from now until 2030.

To estimate the future energy demand from electric vehicles (EVs) and electric cooking, a data-driven curve fitting approach was adopted using both historical energy consumption data and policy-driven future targets. Specifically, cumulative energy demand trajectories were constructed by combining historical data with projected energy values for the year 2030 derived from Nepal's Nationally Determined Contributions (NDC).

For electric cooking, the cumulative energy demand exhibited an approximately linear trend when fitted between the historical observations and the 2030 policy target. Therefore, a linear regression model was employed to describe the growth pattern as shown in Eq. (1):

$$y = 196664x - 4 \times 10^8 \quad (1)$$

Where; x represents the year and y denotes cumulative energy demand.

In contrast, EV energy demand demonstrated a non-linear and rapidly accelerating growth trajectory due to aggressive electrification targets and increasing adoption rates. To capture this behavior, a third-order polynomial function was fitted to the combined dataset (historical values and 2030 target), expressed in Eq. (2):

$$y = 0.0023x^3 - 13.73x^2 + 27722x - 2 \times 10^7 \quad (2)$$

Where; x and y represent the year and cumulative EV energy demand respectively.

Unlike conventional extrapolation approaches, this method constrains the fitted curves to pass through both historical data points and the 2030 NDC-based targets, keeping them consistent with national policy goals while remaining realistic. The resulting functions were then used to estimate annual energy demand trajectories for EVs and electric cooking over the forecast horizon.

2.5. Baseline Demand Forecasting

To isolate intrinsic electricity demand, a baseline demand was computed by subtracting EV and electric cooking energy from total historical demand as shown in Eq. (3):

$$Baseline\ Demand = Total\ Demand - EV\ Energy - Electric\ Cooking\ Energy \quad (3)$$

Subsequently a multivariate LSTM model was created to predict baseline demand using the predicted macroeconomic variables as input data into the LSTM model. The LSTM model utilized Adam Optimizer with MSE loss for training to allow capturing the dependence of multiple influencing variables.

This model predicts the base energy demand using 5 macro-economic features.

- Input Features: 5 (GDP, Population, Urban population (%), Per Capita Energy Consumption, Electricity access).
- Hidden Units: 32
- Architecture:
    - LSTM Layer: 1 layer with 32 hidden units
    - Dropout: 0.2
    - Fully Connected Layer 1: Linear (32 → 16)
    - Activation: ReLU
    - Fully Connected Layer 2: Linear (16 → 1)
- Optimizer: Adam (lr=0.002)
- Training Epochs: 250

2.6. Final Demand Estimation

The total predicted electricity demand was calculated by adding predicted baseline demand to projected EV and electric cooking energy as shown in Eq. (4):

$$Total\ Demand = Baseline\ Demand + EV\ Energy + Electric\ Cooking\ Energy \quad (4)$$

This stacking approach directly incorporates electrification driven demand and provides an all-encompassing estimate of future electricity requirements for Nepal until 2030.

2.7. Model Implementation Details

The models were implemented in Python using the PyTorch framework. The dataset was split chronologically into training (2001–2020) and testing (2021–2025) sets. Model performance was evaluated using RMSE, MAE, and MAPE metrics. Training was performed using full-batch

gradient descent due to the limited dataset size. All preprocessing steps and model configurations are available upon request to ensure reproducibility. The forecast horizon begins in 2026 because the model's hold-out test set spans 2021–2025; the final observed data point (2025) serves as the last calibration anchor, and 2026 therefore represents the first fully out-of-sample year. Selecting 2026 as the base forecast year is consistent with standard forecasting practice in which projections commence immediately after the validation period ends, ensuring that no future observations are used in model training or evaluation.

3. Results and Discussion

3.1. Model Performance and Reliability

A two-step multivariate LSTM model pipeline was developed. The first stage was to use a multivariate LSTM model to predict the macroeconomic drivers of electricity for cooking and electric vehicles. We trained the model with multivariate historical data to predict GDP, total population, and the share of the urban population, and projected induction and EV demand through scenario modelling as exogenous inputs. We then used these estimates to estimate baseline demand for electricity in the second stage.

A chronological split of the dataset (training set from 2001-2020 with 80% of the data; hold-out test set from 2021-2025 with 20% of the data) was used to ensure generalization to future data for the model. This design choice will ensure that the model is evaluated on future data, which has not been used for training, and mimics real-world forecasting conditions for electric cooking and electric mobility.

The predictive performance of the proposed LSTM model was compared to a persistence model and an ARIMA (1,1,0) model. Table 2 compares the predictive results of the three modelling techniques using the test dataset generated from (2021-2025).

Table 2: Comparison of errors between forecasting models

Metric	LSTM Model	Persistence Baseline	ARIMA (1,1,0)
RMSE	0.4858	1.2482	3.9267
MAE	0.4222	1.1903	3.4724
MAPE (%)	4.38	12.46	33.18

The LSTM model achieves a MAPE of 4.38%, demonstrating strong predictive accuracy and improved generalization compared to baseline models; the poor MAPE for ARIMA models of electric demand forecasts is because ARIMA techniques are not equipped to capture the nonlinear phenomena in electricity demand forecasting and are sensitive to small datasets. The improved

performance compared to baseline models highlights the ability of LSTM to capture nonlinear temporal dependencies in electricity demand data.

Final Training Performance; Mean Squared Error (MSE) = 0.0019. Together, the results provide clear evidence that there is no over-fitting of the model during training and that the LSTM model produces a generalizable forecast based upon the MAPE test-set values. The proposed LSTM framework is reliable for forecasting electricity demand and well suited to scenario-based planning for future grid demand in the context of growing electric cooking and electric mobility.

3.2. Forecasted Macro-Economic Drivers

Figures 2 and 3 below provide projected macroeconomic and energy input factors for electricity demand forecasts. The input factors are gross domestic product (GDP), total population, urban population share, electricity consumption per capita, electricity access, induction cooker energy, and electric vehicle (EV) energy.

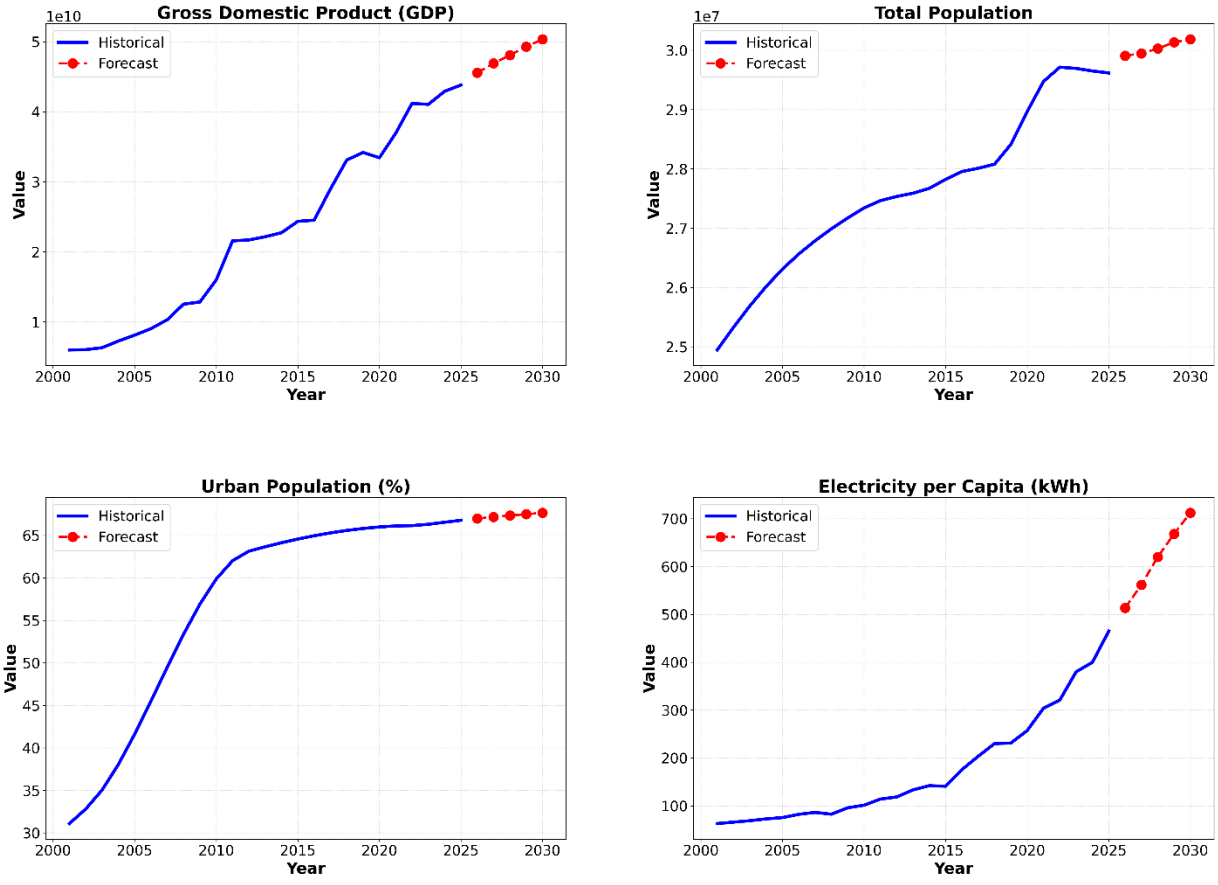


Figure 2: Historical and forecasted trends of key input variables, including GDP, population, urbanization, electricity consumption per capita.

The GDP projection has a stable increase, from approximately 45.6 billion USD by 2026 to approximately 50.3 billion USD by 2030, demonstrating that the economy is growing steadily.

Total population is also slowly increasing from approximately 29.9 million to approximately 30.2 million, indicating that overall, there will be little change to the population demographic during this same time period. However, the urban population share will have a slower but steady rate of growth until it reaches approximately 67.65% by 2030; this suggests that urbanization is progressing to maturity.

Electricity consumption per capita will be significantly increasing from approximately 513 kWh per capita by 2026 to over 712 kWh per capita by 2030; this suggests that the standard of living is expected to rise and the end-use sectors to continue to be electrified at a higher rate.

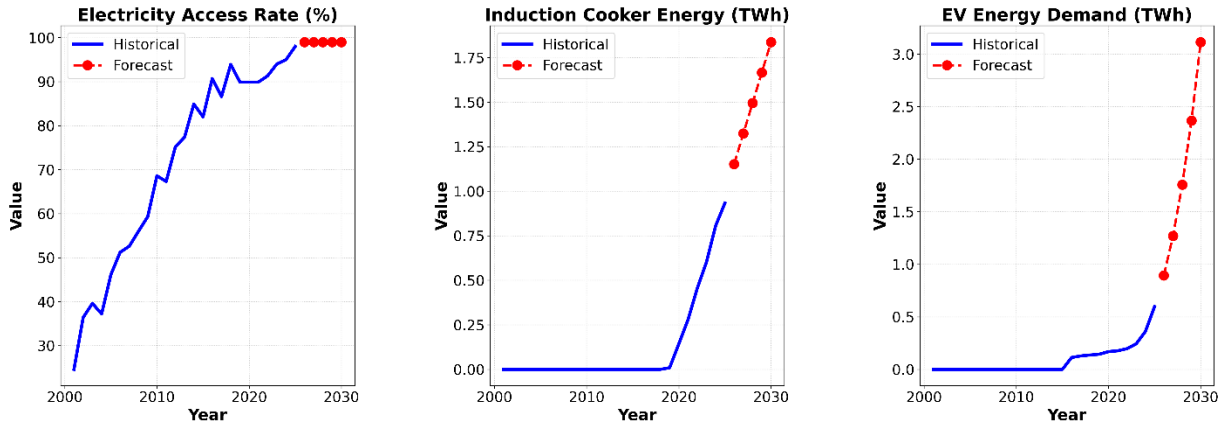


Figure 3: Historical and forecasted trends of key input variables, including electricity access, EV energy, and induction cooker energy.

It is expected that electricity access will be at saturation levels (99%) by 2026 as it aligns with the national goal of electrically connecting the entire nation. Even so, end-use electrification will grow much faster than overall electricity access.

Induction cookers are projected to exhibit a steady increase in energy demand from approximately 1.15 TWh in 2026 to 1.84 TWh in 2030, reflecting the gradual and policy-driven adoption of electric cooking technologies. In contrast, energy demand from electric vehicles (EVs) shows a more rapid and accelerating growth pattern, increasing from approximately 0.91 TWh in 2026 to 3.14 TWh in 2030. This trend highlights the significant impact of transport electrification on future electricity demand, with EVs as a major driver of demand growth.

Taken together, the three projections show that while macroeconomic factors change gradually, the main drivers for growth in future electricity demand will be through the various electrification pathways available, particularly those related to electric mobility and cooking. Rapidly increasing

end-use demand from these two electrification pathways will be a major contributor to shaping future load profiles and how peak demand will be characterized.

3.3. Forecasted Electricity Demand

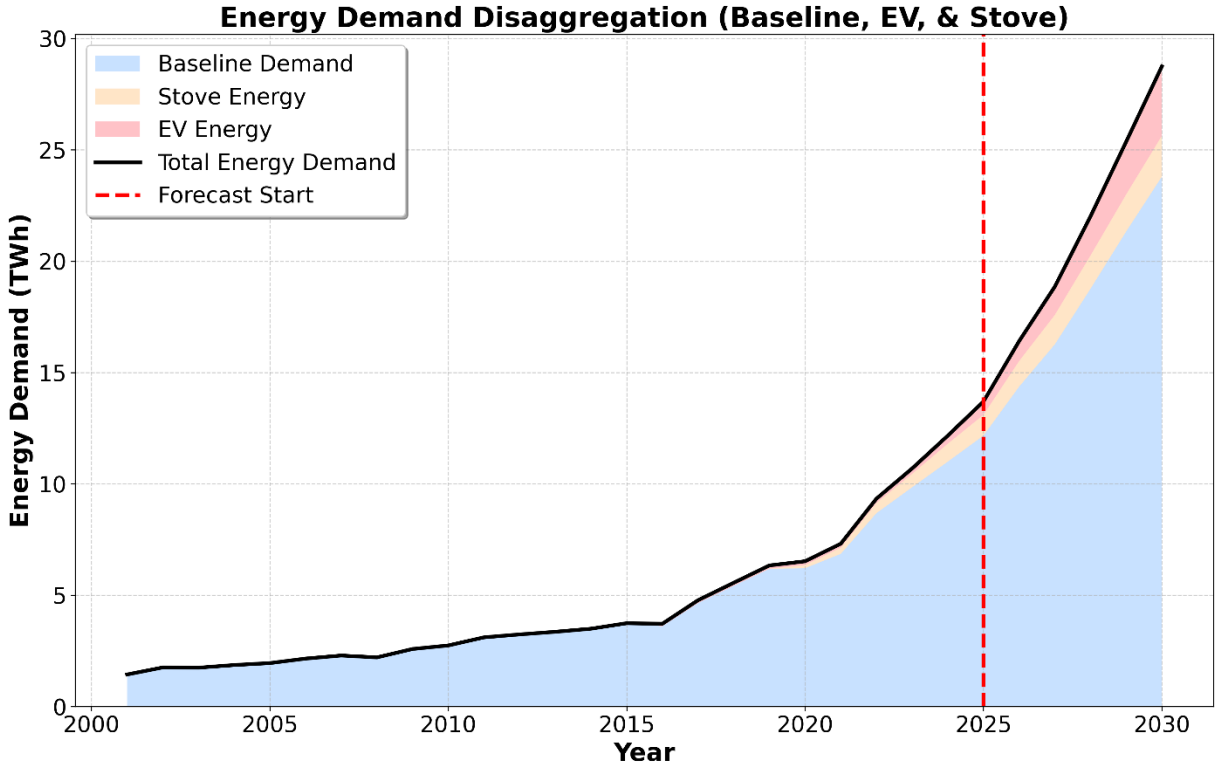


Figure 4: Disaggregation of electricity demand into baseline, EV, and electric cooking components, showing the transition from historical to forecasted demand.

The stacked regions represent baseline demand (bottom layer), electric cooking demand (middle layer), and EV demand (top layer), illustrating their cumulative contribution to total electricity demand. Figure 4 presents the projected electricity demand trajectory from 2001 to 2030, disaggregated into baseline demand, electric cooking (stove) demand, and electric vehicle (EV) demand, with the total demand represented by the black curve. The historical period (2001-2025), shown to the left of the red dashed line, indicates a gradual and relatively stable increase in electricity consumption, rising from approximately 1.5 TWh in 2001 to around 13-14 TWh by 2025. This growth is primarily driven by macroeconomic and demographic factors such as GDP growth, population increase, and improved electricity access, and is captured in the baseline demand component. The curve stays smooth, since no major new demand sources appear in this period.

Beyond 2025, the slope of the total demand curve changes noticeably. While the baseline demand continues its upward trend, the introduction of electric cooking and EV demand adds new layers to the total energy requirement. Electric cooking demand increases from approximately 1.15 TWh

in 2026 to 1.84 TWh in 2030, reflecting a policy-driven and cumulative adoption of induction cookers. This contribution stays fairly linear and predictable because it depends on household adoption rates and steady daily use. In contrast, EV energy demand rises from approximately 0.91 TWh in 2026 to 3.14 TWh in 2030, indicating strong growth in transport electrification. This rapid rise steepens the total demand curve over time.

As a result, total electricity demand grows from 16.42 TWh in 2026 to 28.77 TWh in 2030, indicating nearly a doubling within five years. The widening gap between the baseline layer and the total curve shows electrification, especially in transport, taking on a larger role. By 2030, EV demand alone contributes a substantial share of the incremental increase, surpassing the contribution from electric cooking. So while electric cooking adds a steady, manageable load, EV adoption is the main driver of faster demand growth.

Overall, the figure shows a shift from baseline-driven demand to demand led by electrification. The increasing steepness of the total demand curve after 2025 signals potential challenges for grid planning, particularly in terms of capacity expansion and peak load management. The results emphasize that future energy planning in Nepal must move beyond annual energy adequacy (TWh) and incorporate detailed analysis of demand composition and growth dynamics driven by new electric loads.

3.4. Contribution of EVs and Electric Cooking

Table 3: Forecasted electricity demand in Nepal (2026–2030), including baseline demand and contributions from EVs and electric cooking.

Year	Base Energy	EV Energy	Stove Energy	Total Energy
2026	14.3651	0.9067	1.1523	16.4242
2027	16.2619	1.2867	1.3235	18.8721
2028	18.7597	1.7763	1.4947	22.0307
2029	21.3300	2.3893	1.6659	25.3852
2030	23.7904	3.1395	1.8370	28.7669

The results shown in Figure 4 and Table 3 indicate that one of the most important developments with respect to demand is the rising share of electric vehicles (EVs) and electric cooking. EVs and electric cooking are projected to contribute approximately 4.98 TWh by 2030, representing approximately 17–18% of total electricity demand. EVs account for most of that increase and grow the fastest, making transport electrification the largest new source of demand. Electric cooking grows more slowly but steadily as households shift to cleaner fuels. This marks a shift in what drives Nepal's load: demand once set by baseline economic growth is increasingly driven by technology adoption. Electrification is not just adding to demand; it is changing how electricity is consumed.

3.5. Alignment with Policy Targets (NDC Perspective)

The estimated growth trend correlates in general with Nepal's Nationally Determined Contributions' (NDC) projections. Furthermore, the plans to expand electric mobility and clean cooking align the EV and stove energy trajectories with national policy. The implication of both these energy trajectories counts as an outlook of demand that is based on empirical data as well as the corresponding policies that support it. The projections also indicate that achieving Nepal's electrification goals will require significant, advanced supply-capacity management. Therefore, should these projections of EV and electric cooking take place at a similar pace, substantial upgrades in the generation, transmission, and distribution aspects of the national power grid would be necessary due to the increased load anticipated. Of specific concern is that the projected increase of electricity required for the operation of EVs indicates that the future of transport electrification will be among the highest contributors to national load growth in Nepal.

The distribution composition of the predicted energy usage has a direct impact on peak energy demand, the shape of the 24-hour load curve and estimated load on local distribution networks; electric vehicle (EV) charging and induction cooking have concentrated usage; EVs tend to charge when most vehicles return home in the evening, and induction cookers are used during the morning and evening peak times (more than half of residential peak lighting occurs during this period). From 2026 to 2030, EV demand is estimated to increase from 0.91 TWh to 3.14 TWh which combined with the demand from induction cooking will result in a substantial and more pronounced evening peak load, rather than a gradual increase to baseload. As such, since transformer, feeder and substation sizes are based on estimated peak load, the near-doubling of total demand in five years is likely to significantly underestimate the impact on the distribution infrastructure if charging is not controlled. This emphasizes the importance of demand side management techniques including: time-of-use (TOU) tariffs, smart and managed charging, and load shifting programs to minimize the coincident peak load, as discussed in detail throughout the literature related to the integration of EVs with the electrical grid.

A further complication is Nepal's heavy reliance on hydropower, which is strongly seasonal. The country's generation comes mostly from run-of-river plants, which produce far less energy in the dry winter months, roughly December to April, exactly when heating and cooking demand is highest. In the monsoon season the reverse holds: generation runs to surplus, and much of it is spilled. The electrification trajectories projected here would widen this seasonal supply-demand mismatch: incremental cooking and EV load raises firm winter demand, whereas the available hydro energy to meet it contracts. The present model forecasts demand on an annual basis and does not explicitly resolve this intra-year seasonality or the dispatchability of supply; consequently, the annual adequacy implied by the totals may mask dry-season shortfalls. Addressing this gap will likely require a combination of seasonal storage, dry-season import arrangements, and dispatchable or complementary generation (for example solar, which is comparatively abundant in the dry season) to ensure that electrified cooking and transport loads can be met reliably year-round.

4. Conclusion

In this research, a two-step approach has been proposed using LSTM networks that can forecast electricity usage in Nepal as more people use electricity for their vehicles and their kitchens. The way the above approach operates allows for the easy-to-understand forecasting of how many more kilowatt hours will be consumed in the future by splitting electricity usage into two components: usage that would occur even if no new technology was introduced into the market (baseline usage); and usage that is a result of people transitioning from internal combustion engines (ICE) to electric vehicles (EV), which are driving a majority of incremental usage forecasted as well as people transitioning from using other fuels to electric cooking appliances (both EV and electric cooking usage will continue to grow year over year). By the year 2030, electricity usage resulting from technological improvements such as EVs and electric cooking will account for a large part of total electricity usage, which demonstrates a significant change from electricity demand growing because of standard macroeconomic reasons to demand growing because of technological improvements. These findings illustrate that there will be an immediate need for increased capacity to generate electricity, improvements in the existing transmission and distribution networks, and better demand-side management to support the transition to a cleaner energy future in Nepal.

The framework does have limitations. The EV and electric-cooking projections rest on scenario assumptions and fitted trajectories that may not match real adoption under shifting market and policy conditions, and the historical dataset is small, which can limit how well deep learning models generalize over long horizons. The model also leaves out systemic constraints such as seasonal hydropower variability, peak-load behavior, and operational grid limits. Future work should pair scenario-based analysis with power-system modeling to close these gaps. Even so, the framework is a reliable and flexible tool for forecasting electricity demand in developing countries undergoing rapid electrification, and its results are best read as a medium-term planning estimate rather than an exact prediction.

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